

Project: Meteor Detection via Radio Signals

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Aim

This project aimed to build a system that detects meteors by listening for their unique "ping" in the radio spectrum. Instead of using complex and expensive radar, we created a passive listener. The idea is simple: a distant radio station (our "beacon") is always transmitting. When a meteor burns up in the atmosphere, it leaves a temporary trail of ionized gas. This trail can act like a mirror, briefly reflecting the beacon's signal toward our antenna. Even though the meteor itself might be invisible, we can catch its reflection.

Our goal was to design the antenna, build the data processing pipeline, and create an autonomous system to spot these fleeting signals.

Objectives

Our main objective was to design and implement a cost-effective system that could automatically identify meteors by listening for these reflected radio waves. To break that down, we aimed to:

1. Find a Beacon: Identify a powerful, consistent FM radio station to use as our signal source.
2. Design an Antenna: Create a Yagi-Uda antenna tuned to our beacon's frequency, giving us the needed directionality to focus on a specific patch of sky.
3. Build the Pipeline: Set up the software and hardware to capture and process the radio signal data in real-time.
4. Enable Remote Control: Make the system autonomous and accessible from anywhere.
5. Collect and Analyze Data: Finally, run the system and analyze the data for successful meteor detections.

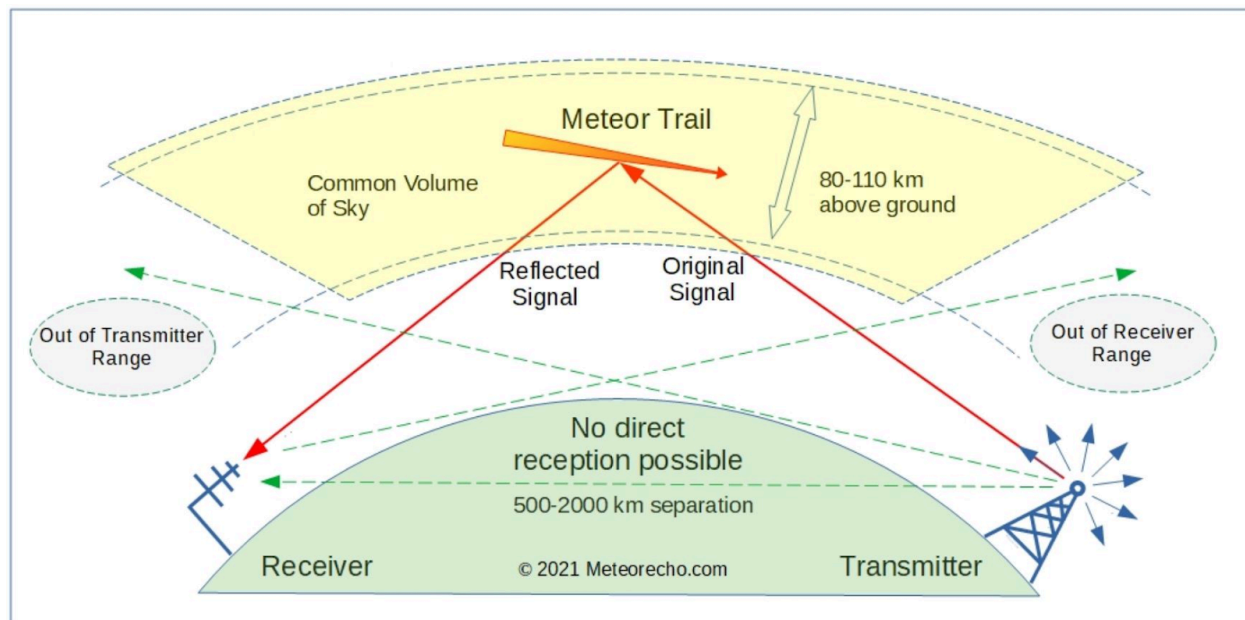
How We Built the System

Tools & Technology:

- Antenna: A 3-element Yagi-Uda antenna design.
- Radio Receiver: An RTL-SDR (RTL2832) dongle to pick up the radio signals.
- Brain & Storage: A Raspberry Pi to run our software, process data, and save the results.

How It Works (Physics):

We point our directional antenna at an empty part of the sky, away from the direct path of our chosen radio station. Normally, we'd hear very little. But when a meteor enters the atmosphere and creates a plasma trail, that trail can reflect the station's signal directly into our antenna for a second or two. Our system is designed to look for that sudden, sharp spike in signal strength—the meteor's signature "ping."



How We Did It (Process):

We started by identifying a suitable beacon, settling on the strong signal from 90.8 FM Nagpur. Next, we designed a 3-element Yagi-Uda antenna specifically for this frequency (the design works well for the 88.9 - 92.6 MHz range). This design gives us the sharp focus we need to ignore interference and listen to a precise point in the sky.

The antenna would connect to the RTL-SDR dongle, which turns the radio waves into digital data. This dongle is plugged into the Raspberry Pi, which acts as the brain of the operation. We wrote Python code for the Pi to continuously monitor the signal strength, looking for the characteristic spikes that indicate a potential meteor

reflection. We also set up remote access (with Blynk IoT) so we could control the system and check its status from our laptops.

Below is the antenna parameter calculations:

```
https://www.changpuak.ch/electronics/yagi_uda_antenna.php
Javascript Version 12.01.2014, based on Rothammel / DL6WU
-----
Frequency      : 90.8  MHz, (useful from 88.984 to 92.616)
Wavelength     : 3304  mm
Rod Diameter   : 10   mm
Boom Diameter  : 20   mm
Boom Length    : 1061  mm
d/lambda       : 0.003   ( min.: 0.002 , max.: 0.01 )
D/lambda       : 0.010   ( min.: 0.01  , max.: 0.05 )
Elements       : 3
Gain           : 5.23 dBd (approx.)
-----
Reflector Length : 1593 mm
Reflector Position : 0 mm
-----
Dipole Position   : 793 mm
-----
Director #1 Position : 1041 mm , Length : 1502 mm
Distance Dipole - Dir. #1 : 248 mm
-----
Directors / Parasitics are isolated.
Please choose an isolater thicker than : 17 mm
```

Below are the parts of the antenna (CAD files) [[Holder1](#)] , [[Holder2](#)]

Objectives Reached

This project laid all the necessary groundwork for successful meteor detection.

Successfully Completed Objectives:

- **Identified a suitable beacon:** 90.8 MHz (Radio 90.8 FM Nagpur).
- **Designed the antenna:** A 3-element Yagi-Uda antenna with specs perfect for our frequency range.
- **Built the data pipeline:** The software pipeline to capture and process data from the SDR is set up and functional.

Objectives Not Reached

- The physical antenna has not been manufactured yet. The 3D model for the antenna has been designed and needs to be 3D printed out and assembled. The electronics have also been bought.
- Because of this, we haven't been able to collect real-world data or perform the final analysis.
- As a result, we have not yet successfully detected a meteor.

Future Work

1. **Manufacture the Antenna:** Build the antenna we've designed.
2. **Collect and Analyze Data:** Deploy the full system and start gathering data. A great target would be during a known meteor shower, like the Perseids, to increase our chances of a detection.
3. **Successfully Detect Meteors:** The ultimate goal—seeing that first confirmed ping on our graphs and knowing we've "heard" a meteor.

Helpful Resources

<https://pysdr.org/content/intro.html>

Rammohan's Design  Meteor_Detection-Radio_Telescope_List_of_Components_Required...

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